Musical examples

The musical examples, including scores and audios, links to the official Analyzing Classical Form website.

Among the https://acf.research.mcgill.ca/supp_list.phpcomplete list of example we look at how the form inference model perform on only a few of them.

Model output

Identifying presentation

example 2-11

🖼️🔊 **Click to Play**

We distinguish between bindings for idea and bindings for formal unit binding. For the presentation, it is the idea that is repeats, while for the period the repeating thing is the formal unit (ABAC)

```
Input surface:
size = 11
  {a ⟦I None⟧}
  {a ⟦IV None⟧}
```

```
Output analysis:
  size=11 compressionRate=1.0
  Seq
    idea = "a"
    parameters = ⟦I None⟧
                 ⟦IV None⟧
```

```
  size=11 compressionRate=1.0
  Concatenate
    {"a" ⟦I None⟧}
    {"a" ⟦IV None⟧}
```

This example illustrates the basics of how variable binding leads to smaller size. By default, we assign a high cost of raw repetition markers from the input (as they are placeholders for the surface motivic ideas). On the other hand, calling a binding variable just cost 1.

example 2-12

🖼️🔊 **Click to Play**

This time we make the input a little bit harder in that the basic idea is not an atomic marker, but the concatenation of two. And the model need to identify the repetition of “a”+“b” as the basic idea, and correctly build up the phrase parameter for the concatenated phrases.

```
Input surface:
size = 27
  {a ⟦I-V None⟧}
  {b ⟦I-III None⟧}
  {a ⟦VI-II None⟧}
  {b ⟦VI-III None⟧}



Output analysis:
  size=25 compressionRate=1.08
  Seq
    idea = "a"+"b"
    parameters = ConcatParam
                   ⟦I-V None⟧
                   ⟦I-III None⟧
                 ConcatParam
                   ⟦VI-II None⟧
                   ⟦VI-III None⟧


  size=27 compressionRate=1.0
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-III None⟧}
      Concatenate
        {"a" ⟦VI-II None⟧}
        {"b" ⟦VI-III None⟧}
```

Identifying compound themes

example 6-1

🖼️🔊 **Click to Play**

```
Input surface:
size = 60
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦IV-V-I None⟧}
  {d ⟦VI-VII-I-V Weak⟧}
  {a ⟦I-V None⟧}
  {b ⟦I-V-I-V Weak⟧}
  {c ⟦V None⟧}
  {e ⟦I Strong⟧}




Output analysis:
  size=59 compressionRate=1.0169491525423728
  Period
    antecedentHead = "a"+"b"
    antecedent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦IV-V-I None⟧}
                     {"d" ⟦VI-VII-I-V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith   $0 ConcatParam   ⟦I-V None⟧ ⟦I-V-I-V Weak⟧
                   Concatenate
                     {"c" ⟦V None⟧}
                     {"e" ⟦I Strong⟧}



  size=61 compressionRate=0.9836065573770492
  Concatenate
    {"a" ⟦I-V None⟧}
    Concatenate
      {"b" ⟦I-V-I-V Weak⟧}
      Period
        antecedentHead = "c"
        antecedent = Concatenate
                       UseIdeaWith   $0 ⟦IV-V-I None⟧
                       Concatenate
                         {"d" ⟦VI-VII-I-V Weak⟧}
                         Concatenate
                           {"a" ⟦I-V None⟧}
                           {"b" ⟦I-V-I-V Weak⟧}
        consequent = Concatenate
                       UseIdeaWith   $0 ⟦V None⟧
                       {"e" ⟦I Strong⟧}
```

example 6-2

🖼️🔊 **Click to Play**

```
Input surface:
size = 64
  {a ⟦I None⟧}
  {b ⟦I None⟧}
  {c ⟦I None⟧}
  {a ⟦V None⟧}
  {b ⟦V None⟧}
  {c ⟦V None⟧}
  {b ⟦I-VI None⟧}
  {b ⟦IV-II None⟧}
  {b ⟦V None⟧}
  {d ⟦I-II-V-I Strong⟧}




Output analysis:
  size=61 compressionRate=1.0491803278688525
  Concatenate
    Seq
      idea = "a"+"b"+"c"
      parameters = ConcatParam
                     ⟦I None⟧
                     ConcatParam
                       ⟦I None⟧
                       ⟦I None⟧
                   ConcatParam
                     ⟦V None⟧
                     ConcatParam
                       ⟦V None⟧
                       ⟦V None⟧
    Sentence
      basicIdea = "b"
      presentation = Seq
                       idea = $0
                       parameters = ⟦I-VI None⟧
                                    ⟦IV-II None⟧
      continuation = Concatenate
                       {$0 ⟦V None⟧}
                       {"d" ⟦I-II-V-I Strong⟧}



  size=60 compressionRate=1.0666666666666667
  Concatenate
    Concatenate
      Seq
        idea = "a"+"b"+"c"
        parameters = ConcatParam
                       ⟦I None⟧
                       ConcatParam
                         ⟦I None⟧
                         ⟦I None⟧
```

```
                ConcatParam
                  ⟦V None⟧
                  ConcatParam
                    ⟦V None⟧
                    ⟦V None⟧
  {"b" ⟦I-VI None⟧}
Concatenate
  Seq
    idea = "b"
    parameters = ⟦IV-II None⟧
                 ⟦V None⟧
  {"d" ⟦I-II-V-I Strong⟧}
```

example 6-5

🖼️🔊 **Click to Play**

```
Input surface:
size = 92
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {d ⟦V Weak⟧}
  {a ⟦I-V None⟧}
  {a ⟦V-I None⟧}
  {b ⟦IV-I None⟧}
  {b ⟦IV-I None⟧}
  {c ⟦II None⟧}
  {c ⟦I None⟧}
  {e ⟦V-I Strong⟧}




Output analysis:
  size=80 compressionRate=1.15
  Period
    antecedentHead = Twice "a"+Twice "b"+Twice "c"
    antecedent = Concatenate
                   UseIdeaWith
                     $0
                     ConcatParam
                       ConcatParam
                         ⟦I-V None⟧
                         ConcatParam
                           ⟦V-I None⟧
                           ⟦IV-I None⟧
                       ConcatParam
                         ⟦IV-I None⟧
                         ConcatParam
                           ⟦II None⟧
                           ⟦I None⟧
                   {"d" ⟦V Weak⟧}
    consequent = Concatenate
                   UseIdeaWith
                     $0
                     ConcatParam
                       ConcatParam
                         ⟦I-V None⟧
                         ConcatParam
                           ⟦V-I None⟧
                           ⟦IV-I None⟧
                       ConcatParam
                         ⟦IV-I None⟧
                         ConcatParam
                           ⟦II None⟧
                           ⟦I None⟧
```

```
                {"e" ⟦V-I Strong⟧}



size=85 compressionRate=1.0823529411764705
Period
  antecedentHead = Twice "a"+Twice "b"
  antecedent = Concatenate
                UseIdeaWith
                  $0
                  ConcatParam
                    ConcatParam
                      ⟦I-V None⟧
                      ⟦V-I None⟧
                    ConcatParam
                      ⟦IV-I None⟧
                      ⟦IV-I None⟧
                Concatenate
                  Seq
                    idea = "c"
                    parameters = ⟦II None⟧
                                 ⟦I None⟧
                  {"d" ⟦V Weak⟧}
  consequent = Concatenate
                UseIdeaWith
                  $0
                  ConcatParam
                    ConcatParam
                      ⟦I-V None⟧
                      ⟦V-I None⟧
                    ConcatParam
                      ⟦IV-I None⟧
                      ⟦IV-I None⟧
                Concatenate
                  Seq
                    idea = "c"
                    parameters = ⟦II None⟧
                                 ⟦I None⟧
                  {"e" ⟦V-I Strong⟧}
```

examples from annotation_pilot

```
Input surface:
size = 335
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {C ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A ⟦I None⟧}
  {B ⟦I None⟧}
  {E ⟦I None⟧}
  {D ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}
  {H ⟦I None⟧}
  {I ⟦I None⟧}
  {J ⟦I None⟧}
  {B ⟦I Weak⟧}
  {K ⟦I None⟧}
  {K ⟦I None⟧}
  {L ⟦I None⟧}
  {M ⟦I Weak⟧}
  {A' ⟦I None⟧}
  {A' ⟦I None⟧}
  {N ⟦I None⟧}
  {G' ⟦I Strong⟧}
  {H' ⟦I None⟧}
```

```
  {H' ⟦I None⟧}
  {F ⟦I None⟧}
  {G ⟦I Strong⟧}



Output analysis:
  size=146 compressionRate=2.2945205479452055
  SimpleBinary
    aPart = Period
              antecedentHead = "A"+"B"
              antecedent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             Concatenate
                               {"C" ⟦I None⟧}
                               {"D" ⟦I Weak⟧}
              consequent = Concatenate
                             UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                             HybridAntCont
                               antecedent = Concatenate
                                              {"E" ⟦I None⟧}
                                              {"D" ⟦I Weak⟧}
                               continuation = Concatenate
                                                TwiceFormalUnit
                                                  {"A'" ⟦I None⟧}
                                                Concatenate
                                                  {"F" ⟦I None⟧}
                                                  {"G" ⟦I Strong⟧}
    bPart = Concatenate
              {"H" ⟦I None⟧}
              Concatenate
                {"I" ⟦I None⟧}
                Concatenate
                  {"J" ⟦I None⟧}
                  Concatenate
                    Concatenate
                      {"B" ⟦I Weak⟧}
                      Concatenate
                        TwiceFormalUnit
                          {"K" ⟦I None⟧}
                        Concatenate
                          {"L" ⟦I None⟧}
                          {"M" ⟦I Weak⟧}
                    Concatenate
                      TwiceFormalUnit
                        {"A'" ⟦I None⟧}
                      Concatenate
                        Concatenate
                          {"N" ⟦I None⟧}
                          {"G'" ⟦I Strong⟧}
                        Concatenate
                          TwiceFormalUnit
                            {"H'" ⟦I None⟧}
                          Concatenate
                            {"F" ⟦I None⟧}
```

```
                          {"G" ⟦I Strong⟧}



size=148 compressionRate=2.2635135135135136
Concatenate
  TwiceFormalUnit
    Period
      antecedentHead = "A"+"B"
      antecedent = Concatenate
                     UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                     Concatenate
                       {"C" ⟦I None⟧}
                       {"D" ⟦I Weak⟧}
      consequent = Concatenate
                     UseIdeaWith   $0 ConcatParam   ⟦I None⟧ ⟦I None⟧
                     HybridAntCont
                       antecedent = Concatenate
                                      {"E" ⟦I None⟧}
                                      {"D" ⟦I Weak⟧}
                       continuation = Concatenate
                                        TwiceFormalUnit
                                          {"A'" ⟦I None⟧}
                                        Concatenate
                                          {"F" ⟦I None⟧}
                                          {"G" ⟦I Strong⟧}
  TwiceFormalUnit
    Concatenate
      {"H" ⟦I None⟧}
      Concatenate
        {"I" ⟦I None⟧}
        Concatenate
          {"J" ⟦I None⟧}
          Concatenate
            Concatenate
              {"B" ⟦I Weak⟧}
              Concatenate
                TwiceFormalUnit
                  {"K" ⟦I None⟧}
                Concatenate
                  {"L" ⟦I None⟧}
                  {"M" ⟦I Weak⟧}
            Concatenate
              TwiceFormalUnit
                {"A'" ⟦I None⟧}
              Concatenate
                Concatenate
                  {"N" ⟦I None⟧}
                  {"G'" ⟦I Strong⟧}
                Concatenate
                  TwiceFormalUnit
                    {"H'" ⟦I None⟧}
                  Concatenate
                    {"F" ⟦I None⟧}
                    {"G" ⟦I Strong⟧}
```

```
Input surface:
size = 623
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
```

```
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {g ⟦I None⟧}
  {k ⟦IV None⟧}
  {g ⟦II None⟧}
  {k ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦I None⟧}
  {g ⟦V None⟧}
  {h ⟦V None⟧}
  {g ⟦I None⟧}
  {h ⟦IV None⟧}
  {i ⟦V None⟧}
  {j ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {a ⟦I None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦I None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}
  {a ⟦I None⟧}
  {a ⟦V None⟧}
  {f ⟦V None⟧}
  {b ⟦V Weak⟧}
  {c ⟦I None⟧}
  {c ⟦I None⟧}
  {d ⟦V None⟧}
  {e ⟦I Strong⟧}



Output analysis:
  size=154 compressionRate=4.045454545454546
```

```
Ternary
  exposition = SimpleBinary
                aPart = HybridAntCont
                          antecedent = Sentence
                                         basicIdea = "a"
                                         presentation = Seq
                                                          idea = $0
                                                          parameters = ⟦I None⟧
                                                                       ⟦V None⟧
                                         continuation = Concatenate
                                                          {$0 ⟦I None⟧}
                                                          {"b" ⟦V Weak⟧}
                                continuation = Concatenate
                                                 TwiceFormalUnit
                                                   {"c" ⟦I None⟧}
                                                 Concatenate
                                                   {"d" ⟦I None⟧}
                                                   {"e" ⟦I Strong⟧}
                bPart = HybridAntCont
                          antecedent = Concatenate
                                         Seq
                                           idea = "a"
                                           parameters = ⟦I None⟧
                                                        ⟦V None⟧
                                         Concatenate
                                           {"f" ⟦V None⟧}
                                           {"b" ⟦V Weak⟧}
                                continuation = Concatenate
                                                 TwiceFormalUnit
                                                   {"c" ⟦I None⟧}
                                                 Concatenate
                                                   {"d" ⟦V None⟧}
                                                   {"e" ⟦I Strong⟧}
  contrastingMiddile = RoundedBinary
                         exposition = Concatenate
                                        Seq
                                          idea = "g"+"h"
                                          parameters = ConcatParam
                                                         ⟦I None⟧
                                                         ⟦I None⟧
                                                       ConcatParam
                                                         ⟦V None⟧
                                                         ⟦V None⟧
                                        Concatenate
                                          Concatenate
                                            {"g" ⟦I None⟧}
                                            {"h" ⟦IV None⟧}
                                          Concatenate
                                            {"i" ⟦V None⟧}
                                            {"j" ⟦I Strong⟧}
                         contrastingMiddile = Seq
                                                idea = "g"+"k"
                                                parameters = ConcatParam
                                                               ⟦I None⟧
                                                               ⟦IV None⟧
                                                             ConcatParam
```

```
                                                                  ⟦II None⟧
                                                                  ⟦V None⟧


size=201 compressionRate=3.099502487562189
Concatenate
  TwiceFormalUnit
    HybridAntCont
      antecedent = Sentence
                     basicIdea = "a"
                     presentation = Seq
                                      idea = $0
                                      parameters = ⟦I None⟧
                                                   ⟦V None⟧
                     continuation = Concatenate
                                      {$0 ⟦I None⟧}
                                      {"b" ⟦V Weak⟧}
      continuation = Concatenate
                       TwiceFormalUnit
                         {"c" ⟦I None⟧}
                       Concatenate
                         {"d" ⟦I None⟧}
                         {"e" ⟦I Strong⟧}
  Ternary
    exposition = TwiceFormalUnit
                   HybridAntCont
                     antecedent = Concatenate
                                    Seq
                                      idea = "a"
                                      parameters = ⟦I None⟧
                                                   ⟦V None⟧
                                    Concatenate
                                      {"f" ⟦V None⟧}
                                      {"b" ⟦V Weak⟧}
                     continuation = Concatenate
                                      TwiceFormalUnit
                                        {"c" ⟦I None⟧}
                                      Concatenate
                                        {"d" ⟦V None⟧}
                                        {"e" ⟦I Strong⟧}
    contrastingMiddile = Concatenate
                           RoundedBinary
                             exposition = Concatenate
                                            Seq
                                              idea = "g"+"h"
                                              parameters = ConcatParam
                                                             ⟦I None⟧
                                                             ⟦I None⟧
                                                           ConcatParam
                                                             ⟦V None⟧
                                                             ⟦V None⟧
                                            Concatenate
                                              Concatenate
                                                {"g" ⟦I None⟧}
                                                {"h" ⟦IV None⟧}
```

```
                                        Concatenate
                                          {"i" ⟦V None⟧}
                                          {"j" ⟦I Strong⟧}
                                  contrastingMiddle = Seq
                                                      idea = "g"+"k"
                                                      parameters = ConcatParam
                                                                     ⟦I None⟧
                                                                     ⟦IV None⟧
                                                                   ConcatParam
                                                                     ⟦II None⟧
                                                                     ⟦V None⟧
                                TwiceFormalUnit
                                  HybridAntCont
                                    antecedent = Sentence
                                                 basicIdea = "a"
                                                 presentation = Seq
                                                                idea = $0
                                                                parameters = ⟦I
None⟧
                                                                                ⟦V
None⟧
                                                 continuation = Concatenate
                                                                {$0 ⟦I None⟧}
                                                                {"b" ⟦V Weak⟧}
                                    continuation = Concatenate
                                                   TwiceFormalUnit
                                                     {"c" ⟦I None⟧}
                                                   Concatenate
                                                     {"d" ⟦I None⟧}
                                                     {"e" ⟦I Strong⟧}
```

Others

supp 6-1

🖼️🔊 **Click to Play**

example 5-11

🖼️🔊 **Click to Play**